

# PADHAISHURU

GATE Preparation Platform

## GATE CS

Computer Science & Engineering

### TREND-BASED MOCK PAPER

— 02 —

|                 |             |
|-----------------|-------------|
| Duration        | 3 Hours     |
| Total Questions | 65          |
| Maximum Marks   | 100         |
| Sections        | 10 subjects |

#### Difficulty Distribution

Easy: 20 questions (31%)

Moderate: 34 questions (52%)

Difficult: 11 questions (17%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.  
These mock papers are constructed using analysis of historical GATE  
PYQ patterns.  
Historical patterns do not guarantee future examination questions.

# GENERAL INSTRUCTIONS

GATE CS — Trend-Based Mock Paper

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## 1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

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## 2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
  - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
  - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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## 3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
  - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
  - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
  - NAT questions carry no negative marking.
  - MSQ questions carry no negative marking.
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## 4. How to Answer

- For MCQ: Select exactly one option.
  - For MSQ: Select one or more options. Partial credit is NOT awarded.
  - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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## 5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

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## 6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

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## 7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
  - The Question Booklet contains multiple pages. Verify the page count.
  - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1.

2M [no -ve]

*[General Aptitude]*

Ravi had \_\_\_\_\_ younger brother who taught at \_\_\_\_\_ university. He was widely regarded as \_\_\_\_\_ honorable man. Select the option with the correct sequence of articles to fill in the blanks.

- (A) a; a; an
- (B) the; an; a
- (C) a; an; a
- (D) an; an; a

Q2.

2M [no -ve]

*[General Aptitude]*

Antonym of 'ephemeral':

Enter your answer here

Q3.

2M [no -ve]

*[General Aptitude]*

Bird : Nest :: Bee : \_\_\_\_\_ Select the correct option to complete the analogy.

Enter your answer here

Q4.

2M [no -ve]

*[General Aptitude]*

For positive non-zero real variables and , if  $\ln + = 1 [\ln() + \ln()]$  ( 2 ) 2 then, the value of + is

Enter your answer here

Q5.

2M [-0.66]

*[General Aptitude]*

Despite his initial hesitation, Rehman's \_\_\_\_\_ to contribute to the success of the project never wavered. Select the most appropriate option to complete the above sentence.

- (A) ambivalence
- (B) satisfaction
- (C) resolve
- (D) revolve

Q6.

2M [-0.66]

*[General Aptitude]*

If ` denotes increasing order of intensity, then the meaning of the words [walk jog sprint] is analogous to [bothered \_\_\_\_\_ daunted]. Which one of the given options is appropriate to fill the blank?

- (A) phased
- (B) phrased
- (C) fazed
- (D) fused

Q7.

2M [no -ve]

*[General Aptitude]*

In a class of 60 students, 30 play cricket, 25 play football, and 15 play both. How many play neither?

Enter your answer here

Q8.

2M [no -ve]

*[General Aptitude]*

In an engineering college of 10,000 students, 1,500 like neither their core branches nor other branches. The number of students who like their core branches is  $\frac{1}{4}$ th of the number of students who like other branches. The number of students who like both their core and other branches is 500. The number of students who like their core branches is

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*Enter your answer here*

Q9.

2M [-0.66]

*[Engineering Mathematics]*

Determinant of 2x2 identity:

- (A) 0
- (B) 1
- (C) 2
- (D) -1

Q10.

2M [-0.66]

*[Engineering Mathematics]*

Probability of at least one head in 3 coin tosses:

- (A)  $\frac{1}{8}$
- (B)  $\frac{3}{8}$
- (C)  $\frac{7}{8}$
- (D)  $\frac{1}{2}$

Q11.

2M [-0.66]

*[Engineering Mathematics]*

Let A be a  $3 \times 3$  matrix with eigenvalues 1, 2, and 3. Then the trace of  $A^2$  is:

- (A) 6
- (B) 7
- (C) 14
- (D) 15

Q12.

2M [no -ve]

*[Engineering Mathematics]*

Mean of binomial  $n=10, p=0.5$ :

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*Enter your answer here*

Q13.

2M [no -ve]

*[Engineering Mathematics]*

Derivative of  $x^3$  at  $x=2$ :

---

*Enter your answer here*

Q14.

2M [-0.66]

*[Programming and Data Structures]*

Infix to postfix uses:

- (A) Queue
- (B) Stack
- (C) Tree
- (D) Graph

Q15.

2M [no -ve]

*[Programming and Data Structures]*

Avg hash lookup:

- (A)  $O(n)$
- (B)  $O(\log n)$
- (C)  $O(1)$
- (D)  $O(n^2)$

Q16.

2M [no -ve]

*[Programming and Data Structures]*

Inorder traversal of BST gives:

---

*Enter your answer here*

Q17.

1M [no -ve]

*[Programming and Data Structures]*

What is the time complexity of inserting at the beginning of a linked list?

---

*Enter your answer here*

Q18.

1M [-0.33]

*[Algorithms]*

Kruskal uses:

- (A) DFS
- (B) BFS
- (C) Union-Find
- (D) Dijkstra

Q19.

1M [no -ve]

*[Algorithms]*

Worst-case time complexity of merge sort:

---

*Enter your answer here*

Q20.

1M [-0.33]

*[Algorithms]*

LCS uses:

- (A) Greedy
- (B) Divide and Conquer
- (C) Dynamic Programming
- (D) Backtracking

Q21.

1M [-0.33]

*[Algorithms]*

Time complexity of binary search:

- (A)  $O(n)$
- (B)  $O(\log n)$
- (C)  $O(n \log n)$
- (D)  $O(1)$

Q22.

2M [-0.66]

*[Algorithms]*

The time complexity of Dijkstra's algorithm using a binary heap is:

- (A)  $O(V^2)$
- (B)  $O(E \log V)$
- (C)  $O(V \log V + E)$
- (D)  $O(V \log V)$

Q23.

1M [-0.33]

*[Algorithms]*

BFS uses:

- (A) Stack
- (B) Queue
- (C) Heap
- (D) Tree

Q24.

1M [no -ve]

*[Algorithms]*

Merge sort space:

---

*Enter your answer here*

Q25.

2M [no -ve]

*[Operating Systems]*

Using 3 frames, which page replacement algorithm would not cause any page fault for reference string 0,1,2,3,0,1,4,0,1,2,3,4?

---

*Enter your answer here*

Q26.

1M [-0.33]

*[Operating Systems]*

Which of the following is NOT a necessary condition for deadlock?

- (A) Mutual Exclusion
- (B) Hold and Wait
- (C) Preemption
- (D) Circular Wait

Q27.

1M [-0.33]

*[Operating Systems]*

Semaphore with value 1 is:

- (A) Counting
- (B) Binary (mutex)
- (C) Monitor
- (D) Lock variable

**Q28.**

1M [-0.33]

*[Operating Systems]*

TLB stands for:

- (A) Translation Lookaside Buffer
- (B) Time Local Buffer
- (C) Total Logic Block
- (D) None

**Q29.**

1M [no -ve]

*[Operating Systems]*

Banker's algorithm is for:

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*Enter your answer here***Q30.**

1M [-0.33]

*[Operating Systems]*

Round-robin uses:

- (A) Time quantum
- (B) Priority
- (C) Arrival time
- (D) Burst time

**Q31.**

1M [-0.33]

*[Operating Systems]*

Inode contains:

- (A) File name
- (B) Metadata
- (C) File data
- (D) Directories

**Q32.**

1M [-0.33]

*[Computer Networks]*

Which protocol operates at the application layer of the TCP/IP model?

- (A) TCP
- (B) UDP
- (C) HTTP
- (D) IP

**Q33.**

1M [no -ve]

*[Computer Networks]*

CSMA/CD is used in:

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*Enter your answer here***Q34.**

1M [-0.33]

*[Computer Networks]*

OSPF is:

- (A) Distance vector
- (B) Link state
- (C) Path vector
- (D) Hybrid

Q35.

1M [no -ve]

*[Computer Networks]*

Protocol to find MAC from IP:

---

*Enter your answer here*

Q36.

1M [-0.33]

*[Computer Networks]*

TCP connection establishment uses:

- (A) 2-way
- (B) 3-way handshake
- (C) 4-way
- (D) 1-way

Q37.

1M [no -ve]

*[Computer Networks]*

Asymmetric encryption:

---

*Enter your answer here*

Q38.

1M [-0.33]

*[Digital Logic]*

Flip-flop stores:

- (A) 1 bit
- (B) 2 bits
- (C) 4 bits
- (D) 8 bits

Q39.

1M [no -ve]

*[Digital Logic]*

Full adder has how many inputs:

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*Enter your answer here*

Q40.

1M [-0.33]

*[Digital Logic]*The number of Boolean functions of  $n$  variables is:

- (A)  $2^n$
- (B)  $2^{(2^n)}$
- (C)  $2^{(n^2)}$
- (D)  $n^2$

Q41.

1M [-0.33]

*[Theory of Computation]*

Which language is regular?

- (A) Equal 0s and 1s
- (B) Strings ending in ab
- (C) Palindromes
- (D)  $a^n b^n$

**Q42.**

1M [-0.33]

*[Theory of Computation]*

PDA recognizes:

- (A) Regular
- (B) Context-free
- (C) Context-sensitive
- (D) Recursive

**Q43.**

1M [-0.33]

*[Theory of Computation]*

Halting problem:

- (A) Decidable
- (B) Undecidable
- (C) Semi-decidable
- (D) None

**Q44.**

1M [-0.33]

*[Compiler Design]*

Lexical analyzer outputs:

- (A) Tokens
- (B) Parse tree
- (C) Intermediate code
- (D) Assembly

**Q45.**

2M [-0.66]

*[CS]*

Consider sending an IP datagram of size 1420 bytes (including 20 bytes of IP header) from a sender to a receiver over a path of two links with a router between them. The first link (sender to router) has an MTU (Maximum Transmission Unit) size of 542 bytes, while the second link (router to receiver) has an MTU size of 360 bytes. The number of fragments that would be delivered at the receiver is \_\_\_\_\_

- (A) A
- (B) B
- (C) C
- (D) D

**Q46.**

1M [-0.33]

*[Compiler Design]*

Lexical analyzer outputs:

- (A) Tokens
- (B) Parse tree
- (C) Intermediate code
- (D) Assembly

**Q47.**

2M [-0.66]

*[CS]*

A person sold two different items at the same price. He made 10% profit in one item, and 10% loss in the other item. In selling these two items, the person made a total of

- (A) 1% profit
- (B) 2% profit
- (C) 1% loss
- (D) 2% loss

Q48.

1M [-0.33]

*[Digital Logic]*

Flip-flop stores:

- (A) 1 bit
- (B) 2 bits
- (C) 4 bits
- (D) 8 bits

Q49.

2M [no -ve]

*[CS]*

Consider the given sequential circuit designed using D-Flip-flops. The circuit is initialized with some value (initial state). The number of distinct states the circuit will go through before returning back to the initial state is \_\_\_\_\_. (Answer in integer) Organising Institute: IIT Roorkee Computer Science and Information Technology (CS1)

- (A) A
- (B) B
- (C) C
- (D) D

Q50.

2M [-0.66]

*[CS]*

Consider the following code snippet using the fork() and wait() system calls. Assume that the code compiles and runs correctly, and that the system calls run successfully without any errors. `int x = 3; while(x > 0) { fork(); printf("hello"); wait(NULL); x--; }` The total number of times the printf statement is executed is \_\_\_\_\_

- (A) A
- (B) B
- (C) C
- (D) D

Q51.

2M [-0.66]

*[CS]*

Let R1 and R2 be two 4-bit registers that store numbers in 2's complement form. For the operation  $R1 + R2$ , which one of the following values of R1 and R2 gives an arithmetic overflow?

- (A)  $R1 = 1011$  and  $R2 = 1110$
- (B)  $R1 = 1100$  and  $R2 = 1010$
- (C)  $R1 = 0011$  and  $R2 = 0100$
- (D)  $R1 = 1001$  and  $R2 = 1111$

Q52.

2M [no -ve]

*[CS]*

Consider the following C program: `#include <stdio.h> int gate (int n) { int d, t, newnum, turn; newnum = turn = 0; t=1; while (n>=t) t *= 10; t /=10; while (t>0) { d = n/t; n = n%t; t /= 10; if (turn) newnum = 10*newnum + d; turn = (turn + 1) % 2; } return newnum; } int main () { printf ("%d", gate(14362)); return 0; }` The value printed by the given C program is \_\_\_\_\_. (Answer in integer) Organising Institute: IIT Roorkee Computer Science and Information Technology (CS1)

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*Enter your answer here*

Q53.

2M [-0.66]

[CS]

The pie charts depict the shares of various power generation technologies in the total electricity generation of a country for the years 2007 and 2023. Year Solar Year 2007 5% 2023 Hydro Coal Solar Coal 30% 35% 20% 20% Gas Hydro Gas 25% 35% 15% Wind Wind 5% 10% The renewable sources of electricity generation consist of Hydro, Solar and Wind. Assuming that the total electricity generated remains the same from 2007 to 2023, what is the percentage increase in the share of the renewable sources of electricity generation over this period?

- (A) 25%
- (B) 50%
- (C) 77.5%
- (D) 62.5%

Q54.

2M [-0.66]

[CS]

Suppose we are given  $n$  keys,  $m$  hash table slots, and two simple uniform hash functions  $h_1$  and  $h_2$ . Further suppose our hashing scheme uses  $h_1$  for the odd keys and  $h_2$  for the even keys. What is the expected number of keys in a slot?

- (A)  $m$
- (B)  $n$
- (C)  $2n$
- (D)  $n$

Q55.

2M [-0.66]

[CS]

Consider a process  $P$  running on a CPU. Which one or more of the following events will always trigger a context switch by the OS that results in process  $P$  moving to a non-running state (e.g., ready, blocked)?

- (A)  $P$  makes a blocking system call to read a block of data from the disk
- (B)  $P$  tries to access a page that is in the swap space, triggering a page fault
- (C) An interrupt is raised by the disk to deliver data requested by some other process
- (D) A timer interrupt is raised by the hardware

Q56.

1M [-0.33]

[General Aptitude]

Despite his initial hesitation, Rehman's \_\_\_\_\_ to contribute to the success of the project never wavered. Select the most appropriate option to complete the above sentence.

- (A) ambivalence
- (B) satisfaction
- (C) resolve
- (D) revolve

Q57.

2M [no -ve]

[CS]

$\mathbb{N} = \{0, 1, 2, 3, \dots\}$  is the set of non-negative integers. Let  $\mathcal{F}$  be the set of functions from  $\mathbb{N}$  to itself. For any two functions,  $f, g \in \mathcal{F}$ , we define  $(f+g)(i) = f(i) + g(i)$  for every number  $i$  in  $\mathbb{N}$ . Which of the following is/are CORRECT about the mathematical structure  $(\mathcal{F}, +)$ ?

Enter your answer here

Q58.

2M [-0.66]

[CS]

The following two signed 2's complement numbers (multiplicand M and multiplier Q) are being multiplied using Booth's algorithm: M: 1100 1101 1110 1101 and Q: 1010 0100 1010 1010 The total number of addition and subtraction operations to be performed is \_\_\_\_\_. (Answer in integer) Organising Institute: IIT Roorkee Computer Science and Information Technology (CS2)

- (A) A
- (B) B
- (C) C
- (D) D

Q59.

2M [no -ve]

[CS]

Sequence the following sentences in a coherent passage. P: This fortuitous geological event generated a colossal amount of energy and heat that resulted in the rocks rising to an average height of 4 km across the contact zone. Q: Thus, the geophysicists tend to think of the Himalayas as an active geological event rather than as a static geological feature. R: The natural process of the cooling of this massive edifice absorbed large quantities of atmospheric carbon dioxide, altering the earth's atmosphere and making it better suited for life. S: Many millennia ago, a breakaway chunk of bedrock from the Antarctic Plate collided with the massive Eurasian Plate.

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Enter your answer here

Q60.

2M [no -ve]

[CS]

The value of the following limit is \_\_\_\_\_.  $\lim_{x \rightarrow 0} \frac{1}{x^2}$

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Enter your answer here

Q61.

1M [-0.33]

[General Aptitude]

If  $x = -1$  for all real values of  $x$ , which one of the following statements is true?

- (A)  $x^2 = 0$
- (B)  $x^2 = 1$
- (C)  $x^2 = 1$ ;  $x = -1$
- (D)  $x^2 = 0$

Q62.

2M [-0.66]

[CS]

Let  $G$  be an edge-weighted undirected graph with positive edge weights. Suppose a positive constant is added to the weight of every edge. Which ONE of the following statements is TRUE about the minimum spanning trees (MSTs) and shortest paths (SPs) in  $G$  before and after the edge weight update?

- (A) Every MST remains an MST, and every SP remains an SP.
- (B) MSTs need not remain MSTs, and every SP remains an SP.
- (C) Every MST remains an MST, and SPs need not remain SPs.
- (D) MSTs need not remain MSTs, and SPs need not remain SPs.

Q63.

2M [no -ve]

[CS]

Consider the following recurrence relation:  $T(n) = T(n-1) + 1$  for  $n > 1$ ,  $T(1) = 1$ . Which one of the following options is CORRECT?

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Enter your answer here

Q64.

2M [no -ve]

[CS]

Let A be a priority queue for maintaining a set of elements. Suppose A is implemented using a max-heap data structure. The operation Extract-Max(A) extracts and deletes the maximum element from A. The operation Insert(A, key) inserts a new element key in A. The properties of a max-heap are preserved at the end of each of these operations. When A contains n elements, which one of the following statements about the worst case running time of these two operations is TRUE?

---

*Enter your answer here*

Q65.

2M [no -ve]

[CS]

Consider the operator precedence and associativity rules for the integer arithmetic operators given in the table below. Operator Precedence Associativity + Highest Left - High Right Medium Right / Low Right The value of the expression  $3 + 1 + 5 \cdot 2 / 7 + 2 - 4 - 7 - 6 / 2$  as per the above rules is \_\_\_\_\_

---

*Enter your answer here*